



Front End Frameworks Assignment 1

Employee Weekly Compensation Calculator TypeScript

1. Specifications

- Create TypeScript that contains the specified interface, parent and child classes as per the class diagram
- The Employee interface specifies the following contract:
 - i. `displayInformation`
 - 1. The implementing classes will implement a method that will display the child employee details
 - ii. `calculateCompensation`
 - 1. The implementing classes will implement this method using business rules specified below
 - iii. `saveEmployee`
 - 1. The implementing classes will implement this method and execute the data validation methods.
- The Employee class will validate:
 - i. The age of the employee must be greater than or equal to 16
 - 1. Display a console message upon failure
 - ii. The rank must be between 1 and five inclusive
 - 1. Display a console message upon failure
 - iii. The SSN must match the pattern: `###-###-###` (Use RegEx or other means)
 - 1. Display a console message upon failure
- The Fulltime Employee Compensation is calculated with the following business rules within the `calculateSalary` method:
 - i. Overtime hours between 1 – 10 are paid at 1.25 times the hourly rate
 - ii. Overtime hours between 11 – 20 are paid at 1.5 times the hourly rate
 - iii. Overtime hours between 21 – 30 are paid at 1.75 times the hourly rate
 - iv. Overtime hours greater than 30 are paid at 2 times the hourly rate ($(\text{Salary}/40 \text{ Hrs}) * \text{OT Hrs} * 2$)
 - v. Overtime is added to the base salary when calculating the total compensation (`calculateCompensation`)
- The Contract Employee Compensation is calculated with the following business rules within the `calculateCompensation` method:
 - i. Overtime is paid out at a rate of 1.5 times the hourly rate for hours greater than 40
 - 1. Example:
 - a. Rate of \$30/hr for the first 40 hours, \$45/hr for all hours beyond 40
- Create a console application which instantiates a Fulltime employee and Contract Employee and:
 - i. Sets all properties
 - ii. Calls the `saveEmployee` and within the `save` executes the validation of Age, Rank and SSN

Ensure the proper commenting for assignments is included at the top of your code. Commit and push regularly



- iii. A valid save state of each employee type will display the details for the fulltime and contract employees by calling `displayInformation`
- iv. Create one employee object that fails the validation checks for Age, Rank and SSN to test the validations

Employee

